

DIABETIC RETINOPATHY SCREENING REPORT

Smart India Hackathon 2026 | Problem 26038 | MathWorks AI Pipeline

Image: IDRiD_046

Date: 09-Sep-2026 13:47

Quality: BORDERLINE (2/3)

1 AI DIAGNOSIS

DIAGNOSIS

Moderate NPDR

ICDR Severity Level 2 / 4 Confidence 50.1%



URGENT REFERRAL — Ophthalmologist consult within 7 days

2 IMAGE QUALITY METRICS

Sharpness (Laplacian Variance)
Threshold: ≥ 50

28.2

FAIL

Brightness (Mean Intensity)
Threshold: 30 – 210

83.6

PASS

Contrast (Std Deviation)
Threshold: ≥ 25

54.8

PASS

3 DETECTED LESION EVIDENCE

Microaneurysms (MA)
Earliest DR sign — small vascular microleaks

9
detected

Hard Exudates (EX)
Lipid deposits — blood-retinal barrier breakdown

21
detected

Hemorrhages (HE)
Deeper vascular wall ruptures

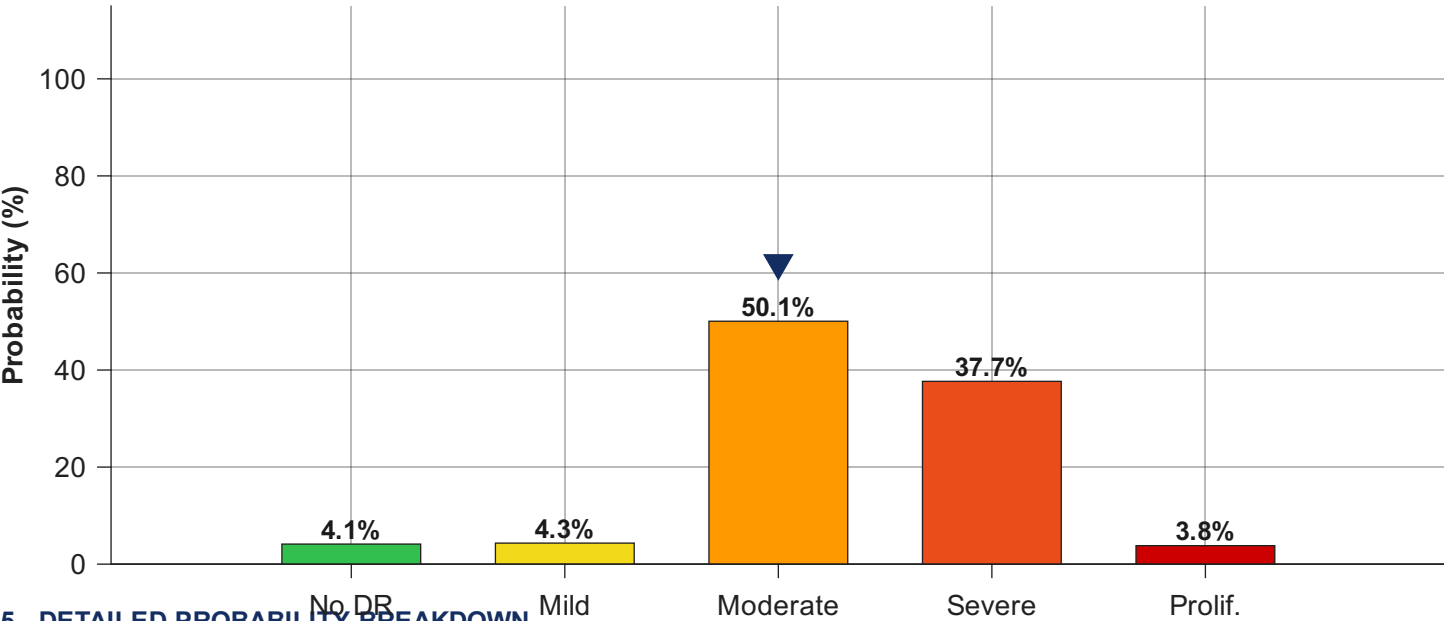
2
detected

Optic Disc [649, 277] Fovea [566, 282]

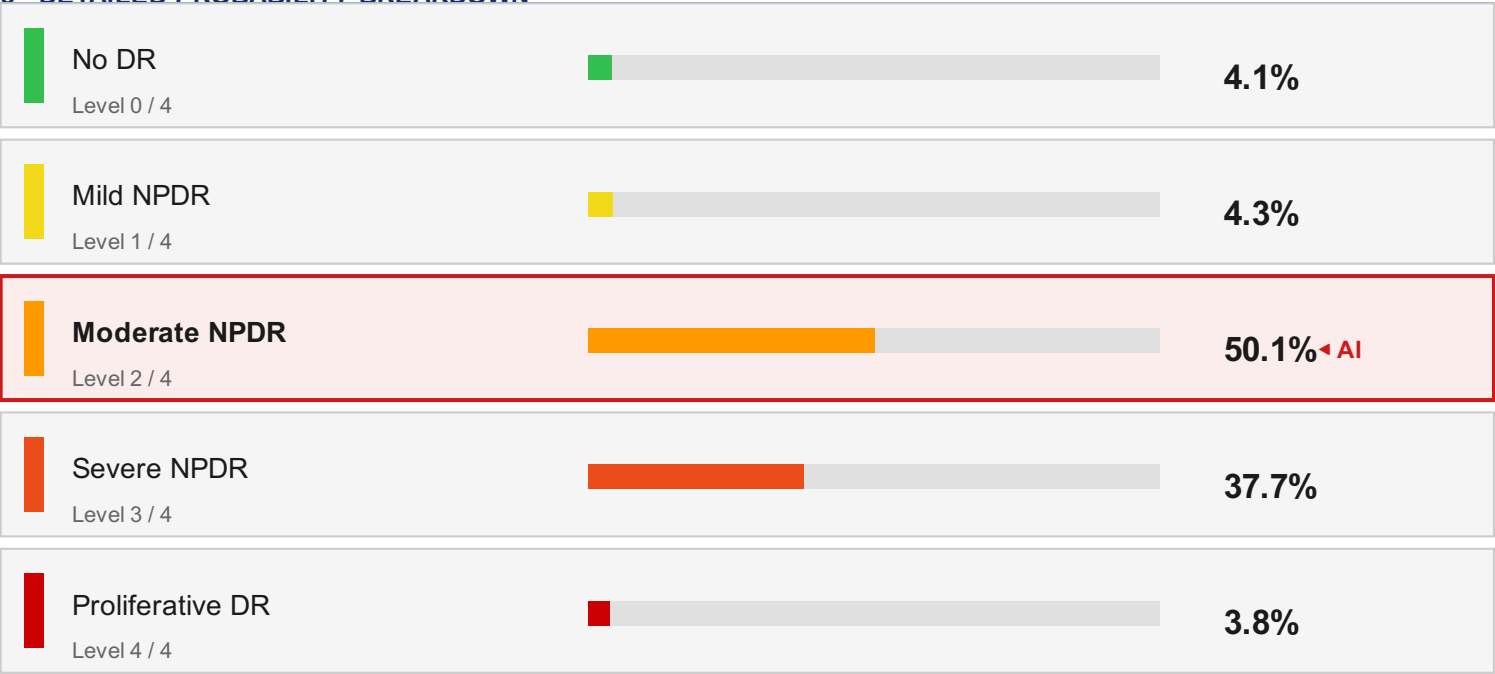
CLASS PROBABILITY ANALYSIS

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4 PROBABILITY DISTRIBUTION



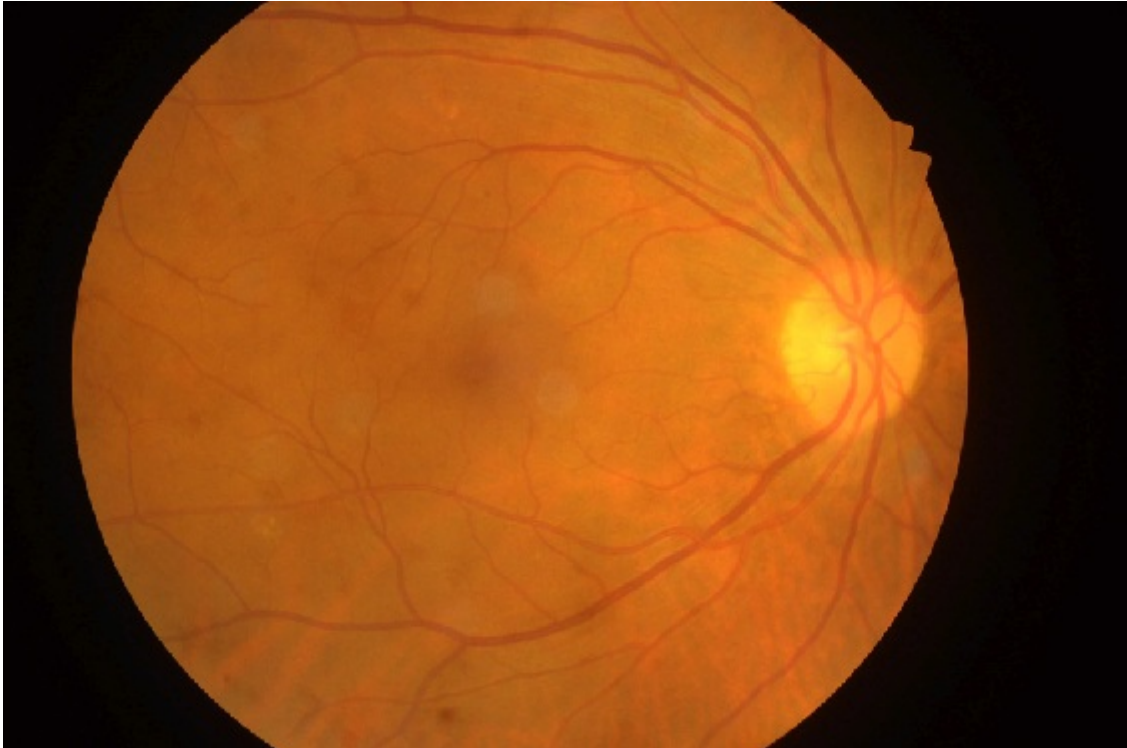
5 DETAILED PROBABILITY BREAKDOWN



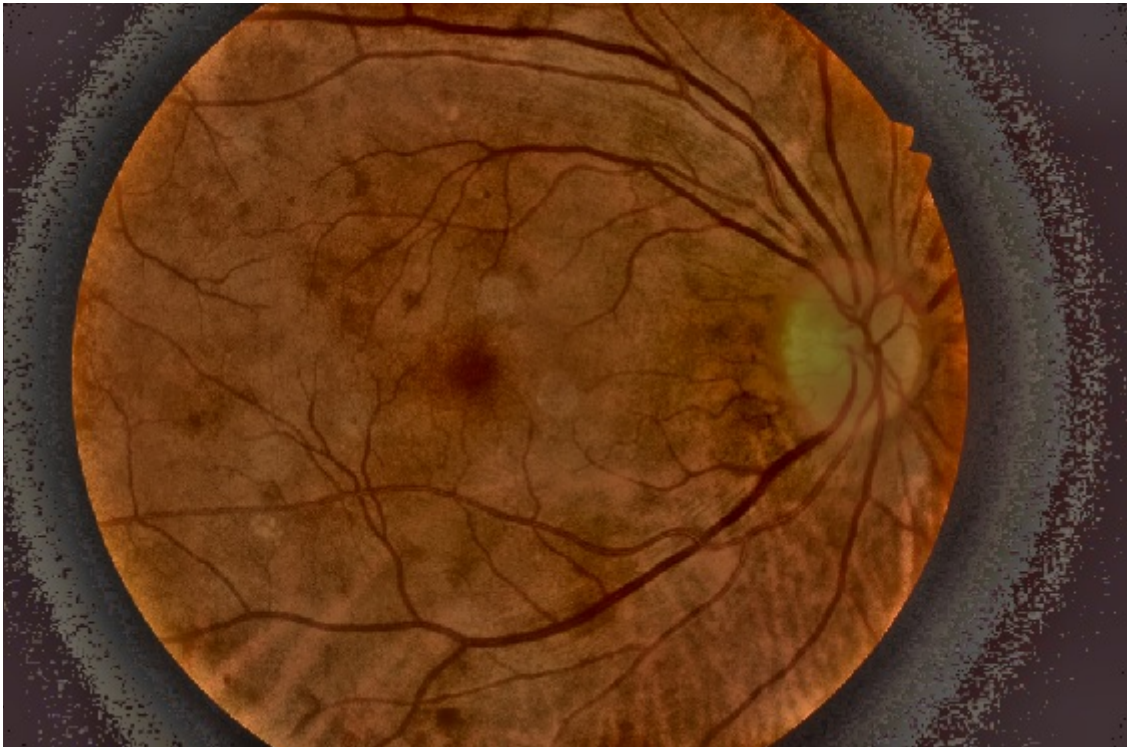
FUNDUS IMAGE QUALITY & ENHANCEMENT

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A ORIGINAL FUNDUS IMAGE



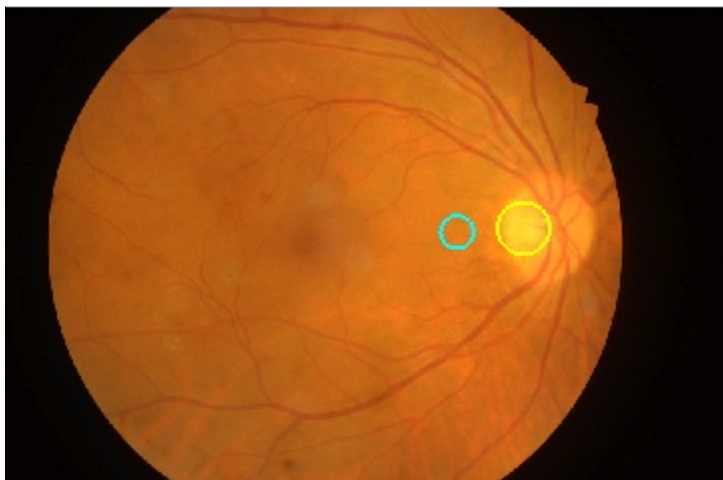
B CLAHE ENHANCED & ILLUMINATION NORMALISED



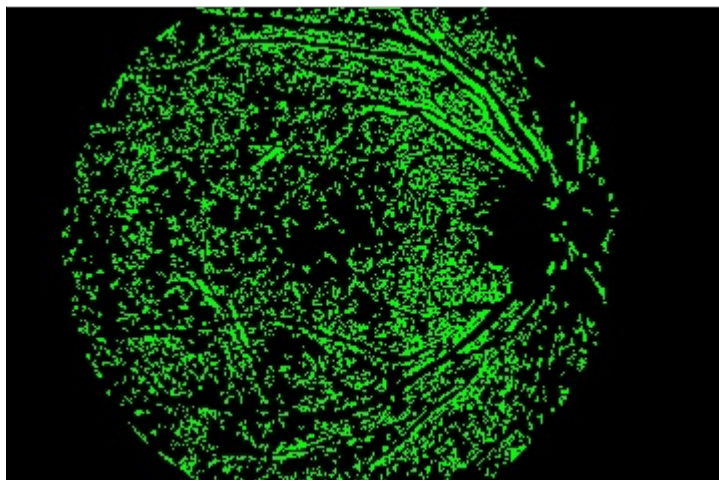
RETINAL SEGMENTATION & EXPLAINABLE AI

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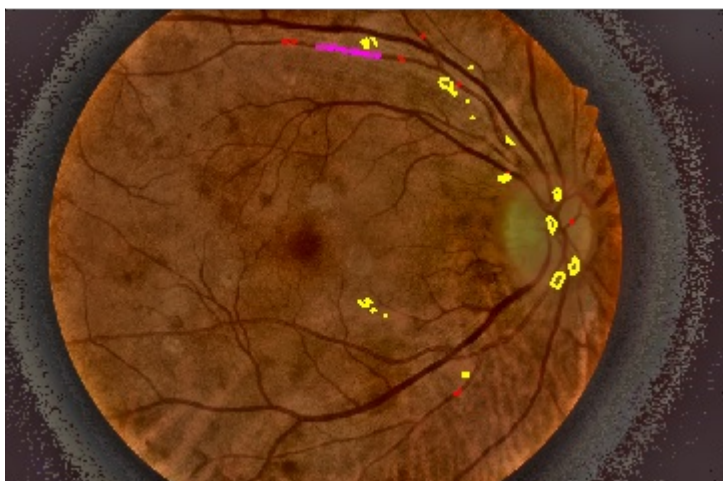
C. Anatomical Landmarks (OD=Yellow, Fovea=Cyan)



D. Vessel Network Segmentation



E. Lesion Map (MA=9, EX=21, HE=2)



F. Grad-CAM Explainability (50.1% Confidence)

